

# DRIVING THROUGH GRAPHS: A BIPARTITE GRAPH FOR TRAFFIC SCENE ANALYSIS

Aditya Humnabadkar<sup>ID</sup>, Arindam Sikdar<sup>ID</sup>, Huaizhong Zhang<sup>ID</sup>, Tanveer Hussain<sup>ID</sup>, Ardhendu Behera<sup>ID</sup>

Department of Computer Science, Edge Hill University, Ormskirk, Lancashire, L39 4QP, UK

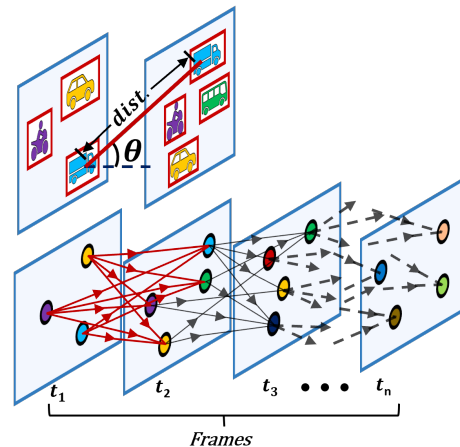
## ABSTRACT

We introduce a novel approach for traffic scene analysis in driving videos by exploring spatio-temporal relationships captured by a temporal frame-to-frame (f2f) bipartite graph, eliminating the need for complex image-level high-dimensional feature extraction. Instead, we rely on object detectors that provide bounding box information. The proposed graph approach efficiently connects objects across frames where nodes represent essential object attributes, and edges signify interactions based on simple spatial metrics such as distance and angles between objects. A key innovation is the integration of dynamic edge attributes, computed using Multilayer Perceptrons (MLP) by exploring this spatial metric. These attributes enhance our Interaction-aware Graph Neural Networks (IA-GNNs) framework by adapting the PageRank-driven approximate personalized propagation of neural predictions (APPNP) scheme and graph attention mechanism in a novel way. This has significantly improved our model's ability to understand spatio-temporal interactions of multiple objects in traffic scenarios. We have rigorously evaluated our approach on two benchmark datasets, METEOR and INTERACTION, demonstrating its accuracy in analyzing traffic scenarios. This streamlined, graph-based strategy marks a significant shift towards more efficient and insightful traffic scene analysis using video data. Our source code is available at: [https://github.com/Addy-1998/Bip\\_DTG](https://github.com/Addy-1998/Bip_DTG).

**Index Terms**— graph neural networks, relational learning, spatio-temporal relationships, knowledge representation, traffic scene analysis, graph attention, and bipartite graphs.

## 1. INTRODUCTION

Driving action/activity recognition using videos/images is an essential task in computer vision and pattern recognition, especially for autonomous vehicle safety and collision avoidance. This technology is crucial in Advanced Driver Assistance Systems (ADAS), where it contributes to developing features like adaptive cruise control and lane-keeping assistance, helping maintain safe distances and alerting drivers to lane changes [1]. The reliability of autonomous vehicles and safety in human-driven scenarios greatly depend on accurately interpreting diverse driving actions. Urban settings, characterized by their dynamic and unpredictable nature [2] with various road users (e.g., vehicles and pedestrians), present significant challenges in driving action recognition. Additionally, the wide range of driving behaviors and



**Fig. 1:** The proposed f2f (frame-to-frame) bipartite graph for modeling multi-object interactions for driving scene recognition by exploring distances and angles among objects.

techniques across different countries [3] necessitates the need for efficient algorithms to accurately recognize and predict actions in these complex scenarios, thereby improving traffic safety and the efficacy of autonomous driving technologies.

Traditional video activity recognition methods have often largely relied on scrutinizing objects [4] and their spatial interactions based on metrics such as distances and angles to model and recognize activities within a video. Such an object-centric approach typically harnesses data like bounding boxes, 3D XYZ coordinates, and keypoints [5, 6] to infer movement and behavior, forming the basis of activity analysis. These methods map out the spatio-temporal relationships between different entities within video frames [7, 8], attempting to construct a narrative of the unfolding actions by interpreting object movements and interactions [9].

However, these traditional methods often struggle in complex scenarios [10] due to their limited representational capacity and the absence of finer details regarding object characteristics necessary for a comprehensive understanding of activities. To overcome these limitations, the field has turned to deep learning, utilizing Convolutional Neural Networks (CNNs) and Long Short-Term Memory Networks (LSTMs) to identify critical spatio-temporal features [11–13]. Nonetheless, these advanced models typically concentrate on video segments [14] rather than entire videos. This segmented analysis may overlook the subtleties of particular actions and does not adapt well to the varying durations of activities, a notable concern in driving scenarios [15]. Furthermore, this approach demands substantial computational

Corresponding author's e-mail id: beheraa@edgehill.ac.uk

resources and may fall short in capturing the intricate interactions in dynamic environments, which are essential for a thorough analysis of traffic behavior [16]. Addressing these limitations, we advocate for a fusion technique that integrates high-level object information with the computational efficiency of GNNs. This synergistic method aims to facilitate a full video representation within manageable computational constraints, obviating the need for labor-intensive feature extraction and providing a more effective framework for traffic activity analysis.

Graph-based action recognition is an innovative approach that leverages graph structures to understand and interpret different object or entity interactions in videos [17]. In the context of traffic scene analysis, this approach is usually achieved using video scene graphs [18]. In this approach, object instances or entities within a scene are represented as nodes in the graph, and edges denote pairwise visual relations or predicates [19]. This approach holds promise in recognizing complex events by capturing spatio-temporal dynamics among objects [20]. However, it presents several challenges. Firstly, constructing accurate and informative graphs from videos can be computationally demanding, especially for lengthy videos. Secondly, the choice of graph structure and algorithms significantly impacts recognition performance. Additionally, scalability becomes an issue when dealing with a large number of objects in a scene. Typically, constructing a video scene graph requires predefined relationships between entities to better understand the context of the action in the video, as highlighted in prior work [21, 22]. However, such predefined relationships may not be readily available for every scene. To address this challenge, our work introduces a dynamic strategy that can automatically infer these relationships as edge features on-the-fly during inference, based on a learned mechanism, thereby contributing to more robust traffic scene analysis. Thus the major contribution of our work can be highlighted as follows:

1. We introduce a novel graph structure for representing videos, utilizing a temporal frame-to-frame (f2f) bipartite graph that effectively models multi-object interactions across adjacent frames. This unique graph structure not only captures complex object interactions but also efficiently represents sequential relationships in longer videos.
2. We introduce a novel fusion approach that combines high-level object information with GNNs to efficiently represent entire videos, significantly enhancing activity recognition in complex traffic scenarios within practical computational limits.
3. We propose an enhanced interaction-aware GNN (IA-GNN) architecture for driving scene analysis. This specialized framework is designed to process bipartite graphs and incorporates dynamic edge attributes from Multilayer Perceptrons (MLPs). The learned MLPs dynamically (during inference) compute edge weights and attributes based on spatio-temporal relationship

metrics, specifically distance and angle measurements between relevant object bounding box centroids that correspond to edges in the temporal f2f bipartite graph. These edge weights and attributes influence the message passing and attention mechanism of our proposed model thereby enabling it to effectively grasp temporal interactions among objects.

In summary, our contributions encompass a graph representation method that utilizes static features such as bounding boxes, and an IA-GNN model. This approach collectively enhances the accuracy and efficiency of traffic scene analysis, representing a significant advancement in the field of driving action recognition. The rest of the paper is organized as follows: Section 2 presents our proposed framework, Section 3 discusses the dataset, implementation details, and performance comparison, and Section 4 concludes our work.

## 2. PROPOSED METHOD

In this section, we formally present the problem definition, temporal f2f bipartite graph construction for modeling the interactions between objects in a driving scene. We then introduce our IA-GNN model for processing these graphs using our dynamic edge weight mechanism.

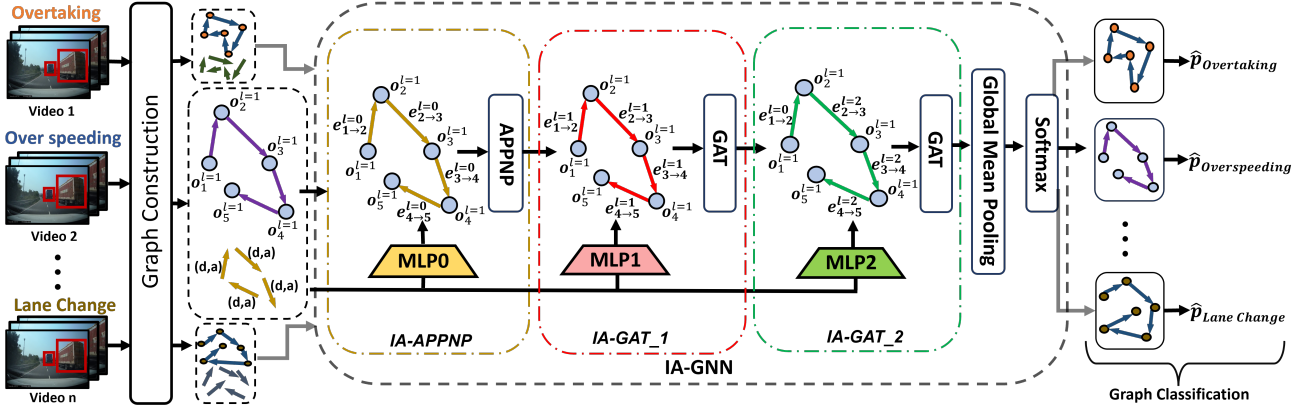
### 2.1. Problem Definition

We introduce the IA-GNN model for classifying events in traffic scenes, utilizing a dataset of  $N$  videos, represented as  $V_n | n = \{1, 2, \dots, N\}$ , each with its corresponding ground truth label  $p_n$ . The model's objective is to predict class probabilities  $\hat{p}_n$  that closely match the actual labels  $p_n$ . Our method consists of two primary components: 1) Creation of a unique temporal f2f bipartite graph, which captures scene dynamics using only object bounding boxes, centroids, and one-hot encoding for object classes. 2) A specifically designed graph-based framework that employs dynamic edge attributes, enhancing scene distinction based on specific object interactions. This approach effectively encapsulates semantic interactions among objects over a scene graph structure.

### 2.2. Temporal Frame-to-Frame (f2f) Bipartite Graph

**Graph Construction.** Given a video  $V_n$  consisting of  $t$  frames, we construct a multi-layer bipartite graph  $\mathcal{G}_n(Q_n, E_n)$  to model temporal interactions among objects within each scene. In this graph, each object is represented as a node. The node set  $Q_n$  corresponds to objects detected across all frames of  $V_n$ , and  $E_n$  represents edges linking objects between adjacent frames (Fig. 1). This structure efficiently encapsulates the sequential relationships observed throughout the video frame sequence  $f_i$ . The objects in frames  $f_1, f_2, \dots, f_k$  are denoted as  $X_1, X_2, \dots, X_k$  respectively, connected in a complete bipartite manner with edges defined by  $E_n = B_{1 \rightarrow 2} \cup B_{2 \rightarrow 3} \cup \dots \cup B_{(k-1) \rightarrow k}$ , linking each set  $X_i$  to the subsequent set  $X_{i+1}$ . The graph is formalized as:

$$\mathcal{G}_n(Q_n, E_n) = G \left( \bigcup_{i=1}^k X_i, \bigcup_{i=1}^{k-1} B_{i \rightarrow (i+1)} \right) \quad (1)$$



**Fig. 2:** Our proposed IA-GNN model consists of the construction of temporal f2f bipartite graphs for modeling spatio-temporal relationships between objects. The MLP0, MLP1 and MLP2 dynamically compute edge weights and attributes based on the distance and angle measurements between the objects. These graphs undergo processing with IA-APPNP for weight adjustments, then through IA-GAT\_1 and IA-GAT\_2 for feature enhancement. Finally, a global mean pooling followed by a Softmax ensures efficient feature aggregation and classification of driving actions.

where,  $G(\cdot, \cdot)$  functions as a graph constructor for any video  $n$ , and  $B_{i \rightarrow (i+1)}$  represents the set of edges connecting nodes  $X_i$  and  $X_{i+1}$ , quantified as  $|B_{i \rightarrow (i+1)}| = |X_i| \times |X_{i+1}|$ . The total count of nodes and edges in this graph are  $\sum_{i=1}^k |X_i|$  and  $\sum_{i=1}^{k-1} |X_i| \times |X_{i+1}|$ , respectively. We uniformly sample frames at regular intervals, currently every 5<sup>th</sup> frame (i.e. every 0.167 second for a 30fps video), to manage the computational demands of graph construction. We also include object types (e.g., cars, buses, cycles, pedestrians, etc.) as a one-hot encoding. We consider every object that appears in a given activity video clip.

**Node Attributes.** Let  $\mathbf{o}_p$  represent the  $p^{th}$  object, serving as an arbitrary node in the set  $\mathcal{Q}_n$  of our graph structure. For each node in  $\mathcal{Q}_n$ , we assign a set of attributes to effectively feed the entire graph into our GNN model. These attributes include three types of object information, namely, 1) Normalized bounding box information,  $\mathbf{b}_p \in \mathbb{R}^4$ , where the first two elements,  $(x, y)$ , are the top-left coordinates of the bounding box, while the last two,  $(w, h)$ , denote its width and height. These values are normalized by the frame's dimension  $W \times H$ , formulated as  $\mathbf{b}_p = [\frac{x}{W}, \frac{y}{H}, \frac{w}{W}, \frac{h}{H}]$ . 2) Normalized Centroid,  $\mathbf{c}_p = [\frac{x_c}{W}, \frac{y_c}{H}] \in \mathbb{R}^2$ , attribute specifies the centroid  $(x_c, y_c)$  of the bounding box. 3) Object type information in the form of 4-bit one-hot encoding,  $\mathbf{e}_p \in \mathbb{R}^4$ . After concatenating these attributes, the final feature for each object instance becomes  $\mathbf{o}_p = [\mathbf{b}_p, \mathbf{c}_p, \mathbf{e}_p] \in \mathbb{R}^{10}$  acting as overall node attribute representation capturing both spatial and categorical characteristics of each object.

**Dynamic Edge Attributes.** In the graph  $\mathcal{G}_n$ , let  $e_{p \rightarrow q} \in B_{i \rightarrow (i+1)}$  denotes an edge that connects the  $p^{th}$  object in frame  $f_i$  to the  $q^{th}$  object in the subsequent frame  $f_{i+1}$ . The attribute of this edge is defined as:

$$e_{p \rightarrow q} = MLP_l([dist(\mathbf{c}_p, \mathbf{c}_q), ang(\mathbf{c}_p, \mathbf{c}_q)]) \quad (2)$$

Here,  $dist(\cdot)$  computes the Euclidean distance, and  $ang(\cdot)$  determines the angle between the centroid vectors  $\mathbf{c}_p$  and  $\mathbf{c}_q$ . The function  $MLP_l(\cdot)$ , linked to the  $l^{th}$  GNN layer, transforms spatial metrics into context-sensitive edge weights and

features. This adaptive approach, specific to each layer, improves the network's ability to analyze temporal interactions between objects in varying frames, capturing the video's dynamic nature.

**Complexity analysis.** The complexity of constructing our  $t$ -layered bipartite graph  $\mathcal{G}_n$  is analyzed by evaluating node and edge creation complexities. For a video  $V_n$  with  $\bar{x}$  average number of objects per frame across  $t$  frames, node creation complexity approximates to  $O(t\bar{x})$ . Edge  $E_n$  creation, involving pairwise object connections between successive frames, results in a complexity of  $O(t\bar{x}^2)$ . The overall complexity, combining both node and edge creation, is therefore  $O(t\bar{x}) + O(t\bar{x}^2) \approx O(t\bar{x}^2)$ , under the assumption  $\bar{x} \gg 1$ . In comparison, constructing a fully connected graph with a similar total of  $t \times \bar{x}$  nodes leads to a complexity of approximately  $O((t\bar{x})^2)$  for large  $t$  and  $\bar{x}$ , primarily due to connections extending beyond consecutive frames. This demonstrates the efficiency of our bipartite graph approach in handling longer videos while maintaining a constant average number of objects per frame. It is noteworthy that if  $\bar{x} \approx 1$ , the complexity of  $\mathcal{G}_n$  simplifies to  $O(t)$ , indicating a dependency more on the number of frames than the number of objects per frame.

### 2.3. Interaction Aware GNN

In our GNN framework, we process temporal bipartite graphs augmented with dynamic edge attributes derived from MLPs, as outlined in Eqn (2). We utilize our proposed Interaction-aware approximate personalized propagation (IA-APPNP) as initial GNN transformation as depicted in Fig. 2. The set of nodes is represented as  $\mathbf{O} = \{\mathbf{o}_p\}$ , with the input feature matrix  $\mathbf{O}^{(0)}$  at iteration  $\tau = 0$  shaped as  $\mathbb{R}^{N \times F}$ , where  $N$  is the number of nodes and  $F$  is the number of features per node. Our proposed IA-APPNP applies a neural network transformation to the initial node features  $\mathbf{O}$ , using a PageRank [23] propagation process incorporating the dynamic edges weights into weighted adjacency matrix. The transformation is governed by:

$$\begin{aligned}
\mathbf{O}^{l(0)} &= \mathbf{O}, \\
\mathbf{O}^{l(\tau)} &= (1 - \gamma) \hat{\mathbf{D}}_w^{-1/2} \hat{\mathbf{A}}_w \hat{\mathbf{D}}_w^{-1/2} \mathbf{O}^{l(\tau-1)} + \gamma \mathbf{O}^{l(0)}, \\
\mathbf{O}^{l+1} &= \mathbf{O}^{l(T)},
\end{aligned} \quad (3)$$

where,  $\gamma \in (0, 1]$  is teleport probability of IA-APPNP and  $\tau$  is the propagation step  $\tau = 0, 1, \dots, T$ . The teleport probability  $\gamma$  enables us to strike a balance between preserving locality (staying close to the root node to avoid over-smoothing) and utilizing information from a broader neighborhood. First, we compute the weighted adjacency matrix  $\mathbf{A}_w$  from the edge weights  $e_{p \rightarrow q}$  using Eqn (2). We then add self-loops to create  $\hat{\mathbf{A}}_w = \mathbf{A}_w + \mathbf{I}$  and construct the diagonal degree matrix  $\hat{\mathbf{D}}_w$ , with each diagonal element  $D_{w,pp} = \sum_{q=0} \hat{A}_{w,pq}$ . The teleport probability  $\alpha$  influences the propagation. After  $T$  iterations,  $\mathbf{O}^{l+1}$  emerges as the updated node features, embodying both local and global graph structures. The normalized adjacency matrix  $\hat{\mathbf{D}}_w^{-1/2} \hat{\mathbf{A}}_w \hat{\mathbf{D}}_w^{-1/2}$  is achieved through symmetric normalization. Here,  $l$  and  $l+1$  denote the input and output, respectively, of a specific ( $l^{th}$ ) GNN layer. Building upon our GNN framework, we developed an Interaction-aware Graph Attention Network (IA-GAT) that integrates a unique dynamic edge-attributes  $e_{p \rightarrow q}$  (see Eqn (2)) as a part of attention coefficients  $\alpha_{p,q}$ , computed as follows:

$$\alpha_{p,q} = \frac{\exp(\text{LReLU}(\mathbf{a}^T [\mathbf{W}\mathbf{o}_p \parallel \mathbf{W}\mathbf{o}_q] + \mathbf{a}_e^T \mathbf{E}_w e_{p \rightarrow q}))}{\sum_{r \in \mathcal{N}(p) \cup \{p\}} \exp(\text{LReLU}(\mathbf{a}^T [\mathbf{W}\mathbf{o}_p \parallel \mathbf{W}\mathbf{o}_r] + \mathbf{a}_e^T \mathbf{E}_w e_{p \rightarrow r}))} \quad (4)$$

Here,  $\parallel$  denotes the concatenation operation,  $\mathbf{o}_p$  and  $\mathbf{o}_q$  are feature vectors of nodes  $p$  and  $q$ ,  $\mathbf{W}$  is the node feature weight matrix, and  $\mathbf{E}_w$  weights the edge attributes.  $\mathbf{a}$  and  $\mathbf{a}_e$  are learnable parameters that integrate node and edge features. The updated feature  $\mathbf{o}_p^{l+1}$  for node  $p$  is obtained by aggregating the attention-weighted features of its neighbors:

$$\mathbf{o}_p^{l+1} = \sigma \left( \sum_{q \in \mathcal{N}(p)} \alpha_{pq} \mathbf{W}\mathbf{o}_q^l \right) \quad (5)$$

This process, including node  $p$  and its neighbors in  $\mathcal{N}(p)$ , along with the nonlinearity introduced by Sigmoid function  $\sigma$ , enriches the feature representation of each node. Finally, we apply global mean pooling to aggregate important node information related to the overall interaction of objects in the video, followed by a softmax activation to predict action labels for the video. Our model effectively captures local connectivity and edge-specific nuances, enhancing the interpretability and efficacy of GNNs, especially in contexts demanding a detailed understanding of graph structure and edge dynamics.

### 3. EXPERIMENTS

In this section, we discuss the datasets used, evaluation measures, and implementation details. We then present a performance comparison with existing state-of-the-art (SOTA) methods, followed by a detailed ablation analysis.

#### 3.1. Dataset Description and Evaluation

The evaluation of our proposed method utilized two key datasets: the Massive Dense & Heterogeneous Behavior

Dataset for Autonomous Driving (METEOR) [24] and The International, Adversarial and Cooperative moTION Dataset (INTERACTION) [25]. The METEOR Dataset encompasses 1250 one-minute videos capturing various Indian traffic scenes. It includes over 2 million annotated frames, providing bounding boxes and trajectories across 16 distinct agent categories, with a total of more than 13 million bounding boxes for traffic agents. Although the original dataset description references 16 goal-oriented actions, annotations and action labels are available for only 10 of these actions, which we used for our experiments. To align with prior research, we utilized 1000 video clips for training and 231 for testing [24]. The INTERACTION Dataset, on the other hand, consists of 991.98 minutes of bird's-eye view traffic footage from the US, featuring bounding box and trajectory data for over 36,054 traffic agents. Our study leveraged trajectory information from 5 different traffic scenarios, consisting of roundabouts, unsignalised intersections, and signalised intersections, etc. The INTERACTION dataset was divided into an 80:20 split, resulting in 25,000 training videos and 6,000 test videos for our analysis.

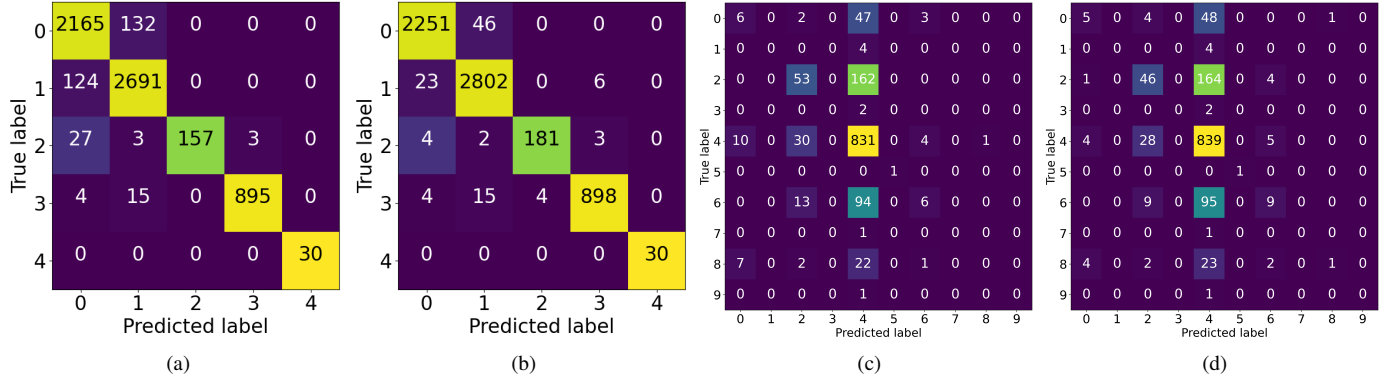
We employed two key metrics: classification accuracy and mean average precision (mAP) to assess the performance.

#### 3.2. Implementation Details

Our implementation utilizes Pytorch 2.12 and Pytorch Geometric 2.1, with each component in Fig. 2 playing a specific role in processing the graph data. The first MLP (MLP0) in IA-APPNP is designed with 2 inputs (distance and angle), 4 hidden states, and a single output, serving to compute the weighted adjacency matrix for APPNP by transforming edge attributes into edge weights. Our IA-APPNP implementation retains the dimensionality of node features at 10, consistent with the dimensions of the initial node inputs. Following this, MLP1, with a structure of 2 inputs, 8 hidden units, and 4 outputs, enriches the first IA-GAT implementation with additional 4-dimensional edge features, which in turn elevates the node feature dimension from 10 to 32. A similar configuration is adopted for MLP2 and the second IA-GAT to further process the graph data. The IA-GAT implementations, each equipped with 4 attention heads, were experimentally set to optimize the efficiency of feature learning. The process concludes with a Global Mean Pooling to aggregate graph data into a singular vector. This vector undergoes refinement via a dense layer before classification with a Softmax layer, which outputs class probabilities, thus efficiently bridging aggregated features to final classification. Our training utilized the Adam optimizer, with experiments conducted at a learning rate of 0.001, a weight decay of 0.5, and a batch size of 8 sequences. The model is trained for 200 epochs. Hyperparameter optimization was guided by extensive experimentation, resulting in the selection of optimal values for  $T = 16$  and  $\gamma = 0.1$ , as substantiated by the analyses presented in the ablation study (Section 3.4).

#### 3.3. Performance Comparison

The performance comparison in Table 1 shows a marked improvement of our approach over the SOTA reported in



**Fig. 3:** Confusion matrices of the METEOR and INTERACTION datasets. Fig. 3a and 3b illustrate the differences in performance without and with MLPs on the INTERACTION dataset, while Fig. 3c and 3d depict the same for the METEOR dataset.

**Table 1:** Comparison with the SOTA for METEOR Dataset and results on the INTERACTION Dataset

| Method        | Dataset      | Features               |          | Acc   | mAP          |
|---------------|--------------|------------------------|----------|-------|--------------|
|               |              | Object                 | Frame    |       |              |
| ACAR-Net [24] | METEOR       | Bounding box           | Slowfast | -     | <b>6.10</b>  |
| Ours          |              | Bounding box attribute |          | 69.61 | <b>28.28</b> |
|               | INTERACTION* | Bounding box attribute |          | 98.50 | <b>99.70</b> |

\* Indicates first reported results on this dataset.

Chandra et al [24]. In their work, they employed the ACAR-Net [26] and achieved an mAP of 6.10%, whereas our model performs significantly better with an mAP of 28.58%. The SOTA method uses Swin-T [27] for object detection and Slowfast [14] for context features considering whole frames within videos. In contrast, our approach constructs a temporal f2f bipartite graph representation of driving actions directly from object bounding box information without relying on any complex CNN feature extraction mechanism. Our IA-GNN model is able to capture this spatio-temporal context of the video action scene and provide nuanced traffic analysis. Our method also performs well on the INTERACTION dataset with an accuracy of 98.50% and mAP of 99.70%. We are the first to report these results as there are no existing methods that classify traffic scenes in this dataset, limiting the extent of direct comparison.

The confusion matrices for the INTERACTION and METEOR datasets demonstrate the substantial benefits of integrating MLPs into the model architecture, markedly improving its classification accuracy for diverse driving actions across complex scenarios (Fig. 3). The temporal f2f bipartite graph aids this enhancement by effectively mapping object relationships over time, providing a detailed spatio-temporal context that MLPs utilize to differentiate between actions. Starting with the INTERACTION dataset (Fig. 3a and 3b), which categorizes actions like ‘roundabout’, ‘unsignalized intersection’, ‘zipper merging’, ‘signaled intersection’, and ‘ramp merging’, we observe from the confusion matrix without MLPs (Fig. 3a) shows a fair accuracy in predicting ‘unsignalized intersection’ with a notable number of correct predictions (2691). However, the model demonstrates a clear struggle with the ‘zipper merging’ action, only predicting it correctly 157 times, which suggests potential deficiencies in the feature set used to represent this complex maneuver or

a lack of sufficient training data for this class. Misclassifications, particularly between ‘roundabout’ and ‘unsignalized intersection’, indicate difficulty in distinguishing between these actions without the nuanced pattern recognition that MLP might provide. Upon introducing the MLPs, there is a marked improvement in the model’s performance (Fig. 3b). The number of correct predictions for ‘unsignalized intersection’ increases to 2802, and though ‘zipper merging’ remains challenging, the model with MLPs shows an enhanced ability to differentiate between ‘roundabout’ and ‘unsignalized intersection’, reducing misclassifications. This improvement suggests that MLP provides a more sophisticated data representation, aiding the model in capturing more complex spatio-temporal relationships.

Moving to the METEOR dataset (Fig. 3c and 3d), which involves actions such as ‘Cutting’, ‘Lane Changing’, ‘Over-Speeding’, ‘TrafficLight\_rulebreak’, ‘WrongLane’, ‘Wrongturn rulebreak’, ‘Yield’, and ‘ZigzagMovement’, the confusion matrix without MLPs (Fig. 3c) shows the model struggles with accurate classification. For instance, it makes more errors in predicting ‘Cutting’ and ‘Lane Changing(m)’ compared to the model with MLPs. The introduction of MLPs (Fig. 3d) results in a noticeable increase in correct predictions for actions like ‘OverTaking’ (from 831 to 839) and ‘WrongLane’ (from 94 to 95), albeit with a slight increase in misclassifications for ‘Cutting’ (from 6 to 5) and ‘Over-Speeding’ (from 47 to 48). However, the model with MLPs still faces challenges with ‘Lane Changing(m)’ (a decrease from 162 to 164) and ‘TrafficLight\_rulebreak’, suggesting that while MLPs aid in general performance, there remains a need for further model refinement or additional data to address these specific complexities.

In conclusion, the novel way to enhance spatiotemporal relationships representing graph edges by incorporating MLPs seems to significantly enhance the model’s ability to discern between different actions across both datasets (Table 2). Its implementation particularly benefits the classification of actions that have subtler distinctions, as seen in the reduced misclassification rates. This improvement points to the value of MLP in providing a deeper level of abstraction, which is critical for complex scenario recognition. Nonetheless, the

persistent difficulty with certain actions indicates that further improvements, such as more advanced feature engineering, increased model complexity, or data augmentation, are required to achieve higher accuracy in the model's predictions for the more intricate and less-represented actions.

### 3.4. Ablation Studies

This section explains the ablation studies conducted where we test the efficacy of MLPs in GNNs, check the impact of attention heads for IA-GAT and perform hyperparameter optimisation to select optimal values for  $T$  and  $\gamma$  for IA-APPNP. **Impact of varying graph configurations.** The ablation study detailed in Table 2 highlights the efficacy of MLPs in GNN models across two datasets, METEOR and INTERACTION. It contrasts two setups: Widely used Fully Connected (FC) and the proposed f2f bipartite. The f2f-bipartite configuration, especially when augmented with MLPs, outperforms the FC configuration, achieving the highest accuracy and mean Average Precision (mAP) on both datasets (METEOR: Acc = 69.61% and mAP = 28.10%; INTERACTION: Acc = 95.73% and mAP = 98.56%). The FC configuration, while straightforward and widely utilized, treats all objects in a video equally without considering the temporal dynamics inherent to traffic scenes. This can lead to a dilution of critical contextual information. The performance boost is credited to MLPs' role in processing dynamic edge attributes, such as distances and angles between traffic scene objects representing graph nodes, enabling real-time updates during inference. These computations afford the GNN a refined understanding of temporal interactions, critical for traffic scene analysis. The MLP-enriched feature representation captures context-sensitive nuances, thereby bolstering the network's predictive accuracy in the complex domain of traffic scenes. The theoretical basis for this improvement lies in the fact that MLPs enable the network to handle edge attributes with greater contextual awareness, aiding the model in capturing temporal dynamics more effectively. The incorporation of MLPs thus not only improves performance metrics but also enhances the interpretability and overall efficacy of the proposed model.

**Impact of attention heads.** Analyzing Table 3, we observe the performance metrics for IA-GAT as it varies the number of attention heads ( $h$ ) across two datasets. For the METEOR dataset, the configuration with four attention heads ( $h=4$ ) yields the highest accuracy of 69.61% while the model with two attention heads ( $h=2$ ) achieves the highest mAP of 28.28% compared to the other configurations. This indicates that while the balance of precision and recall is better with two heads, the overall correct classification rate peaks with four heads. This discrepancy between optimal mAP and accuracy could imply that the model with four heads is better at correctly labeling activities overall, but the model with two heads is more reliable when it comes to the confidence of its predictions across all classes. For the INTERACTION dataset, the highest performance metrics are achieved with four attention heads ( $h=4$ ). This shows that for both datasets, the four-head configuration provides the optimal balance for the model's attention mechanism, enabling it to effectively

capture the relevant features necessary for high performance in driving action recognition tasks.

**Impact of Power iterations vs Teleport probability.** Table 4 presents the results of tuning the parameters  $T$  and  $\gamma$  in the IA-APPNP model on the METEOR Dataset. It suggests that the APPNP model's best performance is consistently achieved at  $\gamma = 0.1$  across all values of  $T$ . There is also a noticeable downward trend in accuracy as the value of  $\gamma$  increases, indicating that lower  $\gamma$  values tend to yield better results in this setup. This could reflect the importance of maintaining a strong influence of the closest neighborhood nodes for a given graph node in the propagation process for this particular dataset.

**Table 2:** Accuracy (Acc in %) and mAP (%) comparison for different graph configurations using METEOR and INTERACTION datasets. FC  $\rightarrow$  Fully Connected; F2F-bip.  $\rightarrow$  Temporal F2F-bipartite. The best results are shown in **bold**.

| Datasets $\rightarrow$ |         | METEOR       |              | INTERACTION  |              |
|------------------------|---------|--------------|--------------|--------------|--------------|
| Config. $\downarrow$   |         | Acc          | mAP          | Acc          | mAP          |
| FC                     | w/o MLP | 48.23        | 16.35        | 43.57        | 67.25        |
|                        | w MLP   | 49.67        | 18.21        | 44.86        | 68.32        |
| F2F-bip.               | w/o MLP | 68.91        | 27.70        | 95.08        | 98.26        |
|                        | w MLP   | <b>69.61</b> | <b>28.10</b> | <b>98.50</b> | <b>99.70</b> |

**Table 3:** Varying the no. of attention heads ( $h$ )

| Datasets    | h=1   |       | h=2   |              | h=4          |              | h=6   |       |
|-------------|-------|-------|-------|--------------|--------------|--------------|-------|-------|
|             | Acc.  | mAP   | Acc.  | mAP          | Acc.         | mAP          | Acc.  | mAP   |
| METEOR      | 68.68 | 27.50 | 69.30 | <b>28.28</b> | <b>69.61</b> | 28.10        | 69.10 | 27.8  |
| INTERACTION | 95.75 | 98.74 | 96.66 | 99.20        | <b>98.50</b> | <b>99.70</b> | 97.50 | 98.90 |

**Table 4:** Impact of Accuracy (Acc in %) on  $T$  vs.  $\gamma$  parameters based on experiments conducted on METEOR dataset

| $T \downarrow$ | $\gamma$     |       |       |       |       |       |       |       |       |
|----------------|--------------|-------|-------|-------|-------|-------|-------|-------|-------|
|                | 0.1          | 0.2   | 0.3   | 0.4   | 0.5   | 0.6   | 0.7   | 0.8   | 0.9   |
| <b>1</b>       | 69.14        | 69.14 | 69.30 | 68.45 | 68.61 | 68.53 | 68.38 | 68.45 | 68.30 |
| <b>4</b>       | 69.37        | 69.07 | 68.15 | 68.84 | 68.63 | 68.56 | 68.32 | 68.44 | 68.29 |
| <b>8</b>       | 68.99        | 68.24 | 68.09 | 67.95 | 67.44 | 67.38 | 67.31 | 67.45 | 67.98 |
| <b>12</b>      | 68.76        | 68.68 | 68.15 | 68.04 | 67.49 | 67.44 | 67.30 | 67.21 | 67.14 |
| <b>16</b>      | <b>69.45</b> | 69.40 | 69.34 | 69.28 | 69.14 | 69.07 | 68.96 | 68.39 | 68.03 |
| <b>20</b>      | 69.22        | 69.04 | 68.96 | 68.88 | 68.72 | 68.68 | 68.50 | 68.39 | 68.28 |

## 4. CONCLUSION

In conclusion, this paper introduces a bipartite graph-driven modeling of multi-object interactions for recognizing traffic scenes in driving videos, deviating from traditional deep learning methods that rely on intricate feature extraction such as object-level and/or scene-level CNN features. By utilizing a novel f2f temporal bipartite graph and an IA-GNN model, our approach simplifies the representation of complex driving scenes and enhances our understanding of multi-object interactions. The efficiency and effectiveness of this model are evident in its ability to accurately recognize traffic scenarios, thus showcasing its potential to advance autonomous vehicle technologies and driver assistance systems.

**Acknowledgement:** This research was partially funded by the UKRI EPSRC project ATRACT (EP/X028631/1): A Trustworthy Robotic Autonomous System for Casualty Triage.

## 5. REFERENCES

- [1] Claudine Badue, R nik Guidolini, et al., “Self-driving cars: A survey,” *Expert Systems with Applications*, vol. 165, pp. 113816, 2021.
- [2] Jingbo Wang, Ye Yuan, Zhengyi Luo, Kevin Xie, et al., “Learning human dynamics in autonomous driving scenarios,” in *ICCV*, 2023, pp. 20796–20806.
- [3] Elena Thomas, Connie McCrudden, Zachary Wharton, and Ardhendu Behera, “Perception of autonomous vehicles by the modern society: a survey,” *IET Intell. Transp. Syst.*, vol. 14, no. 10, pp. 1228–1239, 2020.
- [4] Tal Hassner, “A critical review of action recognition benchmarks,” in *CVPRW*, 2013, pp. 245–250.
- [5] Ardhendu Behera, Matthew Chapman, Anthony G Cohn, and David C Hogg, “Egocentric activity recognition using histograms of oriented pairwise relations,” in *VISAPP*, 2014, vol. 2, pp. 22–30.
- [6] Xiaolong Wang and Abhinav Gupta, “Videos as space-time region graphs,” in *ECCV*, 2018, pp. 399–417.
- [7] Tianlei Jin, Fangtai Guo, Qiwei Meng, Shiqiang Zhu, Xiangming Xi, Wen Wang, Zonghao Mu, and Wei Song, “Fast contextual scene graph generation with unbiased context augmentation,” in *CVPR*, 2023, pp. 6302–6311.
- [8] Jenny Fricke, Christopher Plachetka, and Bernd Rech, “Towards knowledge-based road modeling for automated vehicles: Analysis and concept for incorporating prior knowledge,” in *IV Workshops*, 2021, pp. 49–56.
- [9] Masato Tamura, Hiroki Ohashi, and Tomoaki Yoshinaga, “Qpic: Query-based pairwise human-object interaction detection with image-wide contextual information,” in *CVPR*, 2021, pp. 10410–10419.
- [10] Guyue Hu, Bo Cui, Yuan He, and Shan Yu, “Progressive relation learning for group activity recognition,” in *CVPR*, 2020, pp. 980–989.
- [11] Athma Narayanan, Isht Dwivedi, and Behzad Dariush, “Dynamic traffic scene classification with space-time coherence,” in *ICRA*, 2019, pp. 5629–5635.
- [12] Imen Jegham, Anouar Ben Khalifa, Ihsen Alouani, and Mohamed Ali Mahjoub, “Soft spatial attention-based multimodal driver action recognition using deep learning,” *IEEE Sens. J.*, vol. 21, no. 2, pp. 1918–1925, 2020.
- [13] Apoorva Agarwal, Rishabh Dabral, Arjun Jain, and Ganesh Ramakrishnan, “Skew-robust human-object interactions in videos,” in *WACV*, 2023, pp. 5098–5107.
- [14] Christoph Feichtenhofer, Haoqi Fan, Jitendra Malik, and Kaiming He, “Slowfast networks for video recognition,” in *ICCV*, 2019, pp. 6202–6211.
- [15] Yu Kong and Yun Fu, “Human action recognition and prediction: A survey,” *IJCV*, vol. 130, no. 5, pp. 1366–1401, 2022.
- [16] Madeline Chantry Schiappa, Naman Biyani, Prudvi Kamtam, Shruti Vyas, Hamid Palangi, Vibhav Vineet, and Yogesh S Rawat, “A large-scale robustness analysis of video action recognition models,” in *CVPR*, 2023, pp. 14698–14708.
- [17] Preksha Pareek and Ankit Thakkar, “A survey on video-based human action recognition: recent updates, datasets, challenges, and applications,” *Artif. Intell. Rev.*, vol. 54, pp. 2259–2322, 2021.
- [18] Kaifeng Gao, Long Chen, Yulei Niu, Jian Shao, and Jun Xiao, “Classification-then-grounding: Reformulating video scene graphs as temporal bipartite graphs,” in *CVPR*, 2022, pp. 19497–19506.
- [19] Yangjun Ou, Li Mi, and Zhenzhong Chen, “Object-relation reasoning graph for action recognition,” in *CVPR*, 2022, pp. 20133–20142.
- [20] Wentao Bao, Qi Yu, and Yu Kong, “Uncertainty-based traffic accident anticipation with spatio-temporal relational learning,” in *ACM-MM*, 2020, pp. 2682–2690.
- [21] Arnav Vaibhav Malawade, Shih-Yuan Yu, Brandon Hsu, Deepan Muthirayan, et al., “Spatiotemporal scene-graph embedding for autonomous vehicle collision prediction,” *IEEE Internet Things J.*, vol. 9, no. 12, pp. 9379–9388, 2022.
- [22] Nupur Thakur, Prasanth Sai Gouripeddi, and Baoxin Li, “Graph (graph): A nested graph-based framework for early accident anticipation,” in *WACV*, 2024, pp. 7533–7541.
- [23] Johannes Klicpera, Aleksandar Bojchevski, and Stephan G nnemann, “Predict then propagate: Graph neural networks meet personalized pagerank,” in *ICLR*, 2019.
- [24] Rohan Chandra, Xijun Wang, Mridul Mahajan, et al., “Meteor: A dense, heterogeneous, and unstructured traffic dataset with rare behaviors,” in *ICRA*, 2023, pp. 9169–9175.
- [25] Wei Zhan, Liting Sun, Di Wang, Haojie Shi, et al., “Interaction dataset: An international, adversarial and cooperative motion dataset in interactive driving scenarios with semantic maps,” *arXiv preprint arXiv:1910.03088*, 2019.
- [26] Juntong Pan, Siyu Chen, Mike Zheng Shou, Yu Liu, Jing Shao, and Hongsheng Li, “Actor-context-actor relation network for spatio-temporal action localization,” in *CVPR*, 2021, pp. 464–474.
- [27] Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo, “Swin transformer: Hierarchical vision transformer using shifted windows,” in *ICCV*, 2021, pp. 10012–10022.